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MLPerf Storage 2.0 Closed Remotely-Attached Block Device OpenFlex Data24 4200 with 3 clients.

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Hardware

Data24_EBOF_24Drive_3Node

- Client Servers:
  - SuperMicro H12DSU-iN Qty 3
  - Form Factor: 1U
  - CPU: 2x AMD EPYC 7542 (32 cores each)
  - Memory: 512 GB (16x32GB DDR4)
  - OS: Ubuntu 24.04
  - Data Network: Nvidia CX6
- Disaggregated Storage:
  - OpenFlex Data24 4200 Qty 1
  - Form Factor: 2U
  - 24 x Kioxia CM7-V Drives 6.4 TB
- Switch:
  - Mellanox SN3800 64-Port 100GbE QSFP28 with Onyx OS

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Software and Settings
Data24_EBOF_24Drive_3Node

OpenFlex Data24 EBOF

- NVMe Protocol: RDMA

Switch

- Congestion Control: PFC

Client Server

- OS: Ubuntu 24.04
- Kernel: 6.8.0-63-generic
- OFED version: 24.10-1.1.4.0-ubuntu24.04
- Hyperthreads: disabled
- Turbo: enabled
- Sysctl settings:
  - `vm.swappiness=0`
  - `vm.dirty_bytes=50000000`
  - `dev.raid.speed_limit_max=5000000`
  - `dev.raid.speed_limit_min=50000`

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Nameplate Power

Data24_EBOF_24Drive_3Node

Category	Component	Quantity	PSU Nameplate Power	Design Power
Storage	Data24 4200	1	2 x 800W	800W
			Total Power	800W

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Topology

Data24 With 3 Clients

Client Servers 1-3:

- SuperMicro: H12DSU-iN qty 3
- Form Factor: 1U
- CPU: 2x AMD EPYC 7542 (32 cores each)
- Memory: 512 GB (16x32GB DDR4)
- OS: Ubuntu 24.04
- Data Network: Nvidia CX6

